

ONLINE BOOKING SYSTEM

Maestro Autoworks

Presented By:

Yzabelle David

Clyde Ezikel Ilarde

Patrick John Rey Dalupang

System Analysis and Design

Professor

Prof. Genesis Reb Lobarbio

Project Overview

Maestro Autoworks, a car repair shop currently relying on a manual booking system, faces operational inefficiencies that result in disorganized scheduling and an inconvenient experience for both clients and staff, as walk-in customers are subject to uncertain availability without any structured appointment process. To address these challenges, a website will be developed to serve as a comprehensive digital platform where customers can learn about the shop, explore its range of services and history, and book appointments online at their convenience, ultimately streamlining operations, improving customer experience, and establishing a more organized and efficient management system for the business.

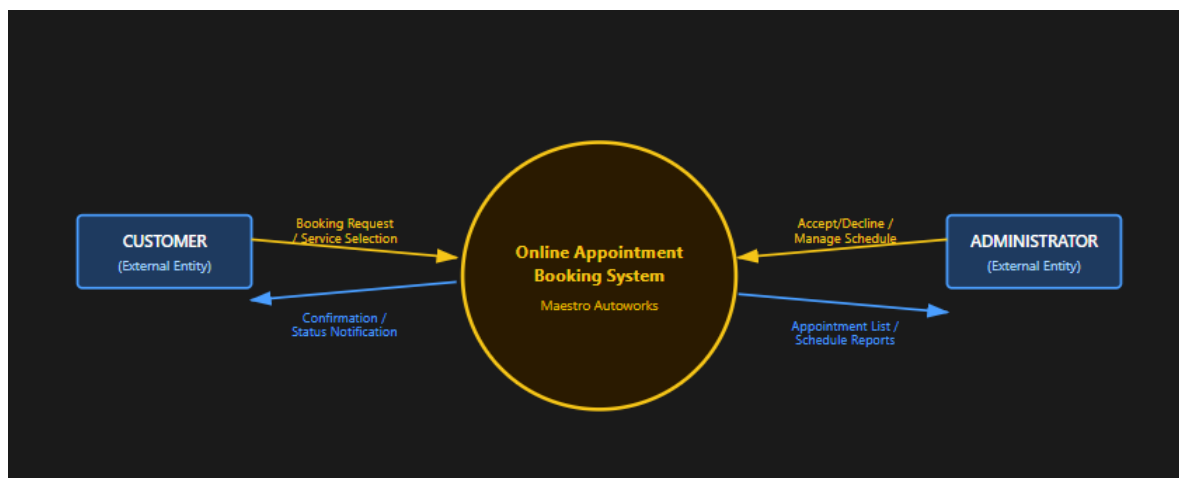
Problem Statement

Maestro Autoworks currently operates without a modernized business system, leaving it dependent on manual and inefficient processes that hinder overall productivity. The absence of a consistent promotional strategy has resulted in minimal business exposure, difficulty attracting new customers, and limited customer retention — despite the shop's advantageous physical location. The existing manual booking system is a significant pain point, as frequent overlapping appointments make it difficult for staff to manage daily workloads effectively and complete vehicle repairs within their designated timeframes. These compounding issues highlight the urgent need for a structured, digital solution that can streamline operations and support the business's growth.

System Objective

The primary objective of this project is to develop and deploy an online appointment booking system that addresses the current operational needs of Maestro Autoworks by providing a centralized and automated platform that improves service efficiency, enhances customer experience, and strengthens the shop's online presence — replacing its reliance on manual and informal scheduling methods with a more organized and professional solution. Specifically, the system aims to provide an easy-to-use booking interface that allows customers to book appointments without having to physically visit the shop, ultimately improving overall operational efficiency and customer convenience.

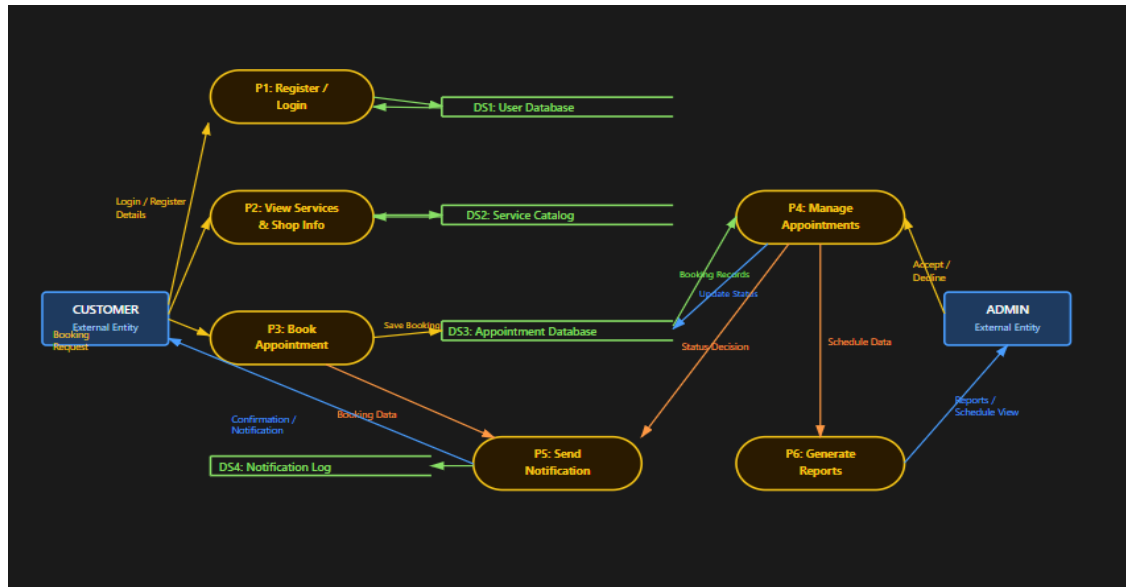
Data Flow Diagram



Maestro Autoworks Online Appointment Booking System, illustrating its boundaries and its interaction with two external entities: the Customer and the Administrator. On the customer side, data flows into the system in the form of booking requests and service selections, while the system responds by sending back appointment confirmations and status notifications. On the administrator side, inputs such as acceptance or decline decisions and schedule management actions are fed into the system,

which in turn generates appointment lists and schedule reports for the administrator's reference. This diagram establishes the overall scope of the system and defines what information enters and exits its boundaries.

DATA FLOW DIAGRAM:



Data Flow Diagram of Maestro expands the context diagram by breaking down the system into six core processes that work together to facilitate the online appointment booking operations of Maestro Autoworks. The first process handles customer registration and login, storing and retrieving account data from the User Database. The second process allows customers to view available services and shop information sourced from the Service Catalog. The third process manages the submission of booking requests, saving appointment records into the Appointment Database. The fourth process is handled by the administrator, who reviews and either accepts or declines bookings, updating the appointment status accordingly. The fifth process is responsible for sending notifications and confirmations back to the customer, logging all sent communications in the Notification Log. Finally, the

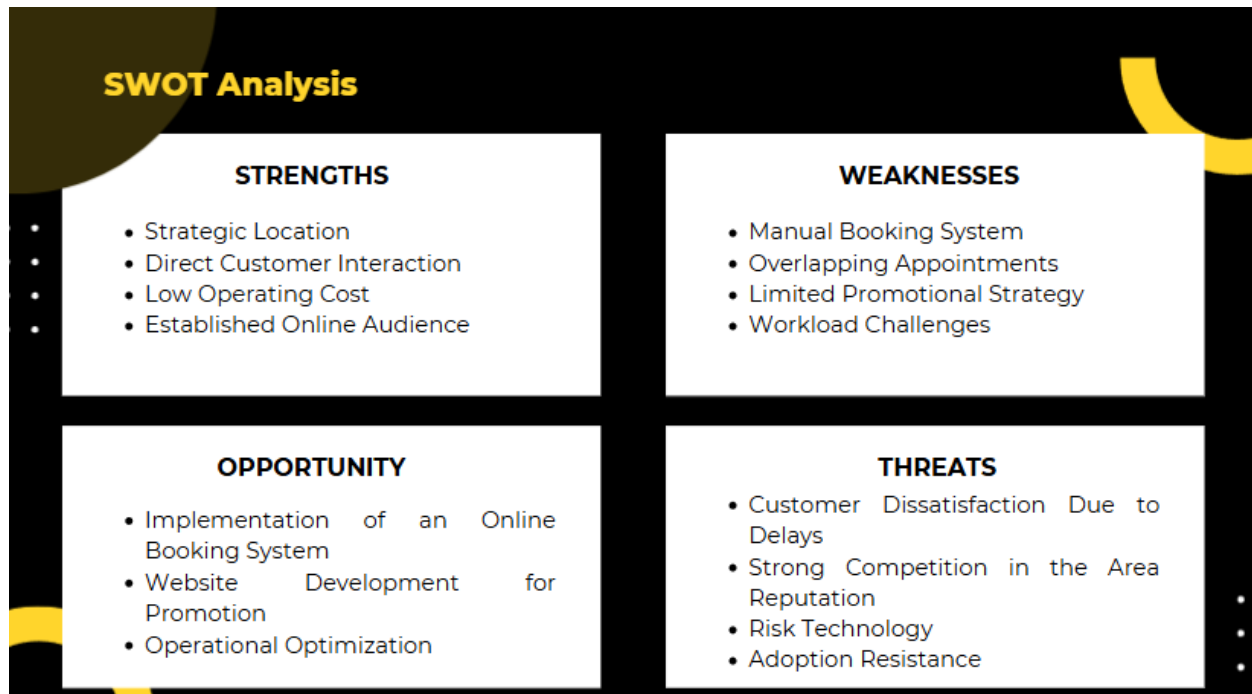
sixth process generates schedule and workload reports for the administrator, supporting more informed decision-making and operational planning.

Scope and Limitation

The Maestro Autoworks Online Appointment Booking System is designed to serve two primary users: administrators and customers. On the administrative side, the system allows authorized personnel to manage and monitor appointments, view and update shop schedules, accept or decline incoming booking requests, and manage service listings to keep the platform's offerings current and accurate. On the customer side, the system enables users to view shop information, services, and history, book appointments online at their convenience, and receive appointment confirmations once their bookings have been processed. Together, these functionalities define the operational boundaries of the system and the range of services it is built to support.

Despite its capabilities, the system operates within certain constraints that define what it is not designed to handle. It does not support real-time vehicle tracking or provide live repair status updates to customers during the service process. Mechanic assignment is also outside the system's scope, as this responsibility remains with the head mechanic and is handled manually. Additionally, the system is limited to accommodating a maximum capacity of five vehicles at a time, reflecting the shop's current physical capacity. The platform requires a stable internet connection to function and is not available as a mobile application, meaning it is only accessible through a web browser on compatible devices.

SWOT ANALYSIS



Maestro Autoworks holds several internal advantages that position it well within its market. Its strategic location provides strong visibility and accessibility to potential customers in the area, while its direct customer interaction allows the shop to build personal relationships and trust with its clientele. The business also benefits from low operating costs, enabling it to remain competitive in its pricing. Furthermore, the shop has already established an online audience, giving it an existing digital presence that can be leveraged to support further growth and customer engagement.

Despite its strengths, Maestro Autoworks faces a number of internal challenges that hinder its operational effectiveness. The shop currently relies on a manual booking system, which is inefficient and prone to errors. This contributes to frequent overlapping appointments that disrupt daily workflow and affect service quality. The absence of a consistent and structured promotional strategy has also limited the shop's ability to reach and retain customers. Combined with workload challenges stemming from

poor scheduling, these weaknesses highlight the need for a more systematic and modernized approach to business operations.

The current gaps in Maestro Autoworks' operations present several opportunities for meaningful improvement and growth. The implementation of an online booking system would directly address the inefficiencies of the existing manual process, streamlining appointment management for both staff and customers. Developing a dedicated website would also serve as an effective promotional tool, increasing brand visibility and attracting a wider customer base. Additionally, optimizing overall operations through digital solutions presents an opportunity to improve service delivery, reduce workload conflicts, and strengthen the shop's competitive position in the local market.

Maestro Autoworks also faces a number of external threats that could impact its sustainability and reputation. Continued delays and scheduling inefficiencies risk causing customer dissatisfaction, which may lead to negative reviews and loss of repeat business. The shop also operates in an area with strong competition, meaning that any decline in service quality could easily drive customers toward rival establishments. The adoption of new technology introduces its own risks, as staff unfamiliarity or resistance to digital tools may slow down implementation and reduce the system's effectiveness. These threats underscore the importance of a well-planned and properly supported transition to a modernized business system.